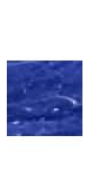
```
Arguments used:
  folder: C:\Users\pensf\Documents\AutomatedLBP_Benchmarking\LBP_Test_Images\Swatches\LBP_Texture_Swatches
  P: 8
  R: 1.0
  method: ror
  metric: cosine
  save_csv: matches.csv
  blur: 1.25
  ltp: True
  visualize: True
  X: 64
  Y: 64
  V: True
  G: False
Correct matches: 47/75 (62.67%)
Highest distance among correct matches: 0.008696
Lowest distance among correct matches: 0.000123
Highest distance among incorrect matches: 0.035644
Lowest distance among incorrect matches: 0.000771
```

```
Index: 0
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.001405
```



```
Index: 1
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.000875
```

```
Index: 2
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.002791
```

```
Index: 3
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.002791
```

```
Index: 4
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.000875
```

```
Index: 5
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.002743
```

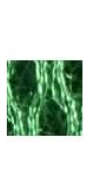
```
Index: 6
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.007214
```

```
Index: 7
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.001365
```

```
Index: 8
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.006497
```

```
Index: 9
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.000771
```

```
Index: 10
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: brown
Distance Metric: Cosine
Distance: 0.001084
```



```
Index: 11
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Cosine
Distance: 0.001396
```

```
Index: 12
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.002342
```

```
Index: 13
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Cosine
Distance: 0.001396
```

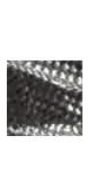
```
Index: 14
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Cosine
Distance: 0.002094
```

```
Index: 15
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Cosine
Distance: 0.002094
```

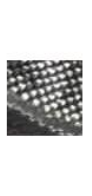
```
Index: 16
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.004148
```

```
Index: 17
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: gray
Distance Metric: Cosine
Distance: 0.004033
```

```
Index: 18
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.009469
```

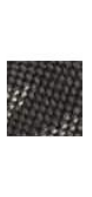


```
Index: 19
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.009840
```



```
Index: 20
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Cosine
Distance: 0.029234
```

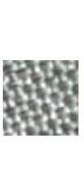
```
Index: 21
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.004241
```



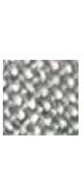
```
Index: 22
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.035644
```

```
Index: 23
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Cosine
Distance: 0.005479
```

```
Index: 24
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Cosine
Distance: 0.002460
```



```
Index: 25
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Cosine
Distance: 0.005479
```



```
Index: 26
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Cosine
Distance: 0.005157
```

```
Index: 27
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Cosine
Distance: 0.002257
```

```
Index: 28
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Cosine
Distance: 0.002257
```

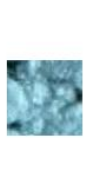
```
Index: 29
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.001383
```

```
Index: 30
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.001500
```

```
Index: 31
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.002505
```

```
Index: 32
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.002549
```

```
Index: 33
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.001383
```



```
Index: 34
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.001565
```

```
Index: 35
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.001299
```

```
Index: 36
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.004033
```

```
Index: 37
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.001866
```

```
Index: 38
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: gray
Distance Metric: Cosine
Distance: 0.005275
```

```
Index: 39
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.002845
```

```
Index: 40
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.006131
```

```
Index: 41
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.001236
```

```
Index: 42
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.002173
```

```
Index: 43
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.002070
```

```
Index: 44
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.002173
```

```
Index: 45
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.001405
```

```
Index: 46
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.002486
```

```
Index: 47
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.000747
```

```
Index: 48
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.002657
```

```
Index: 49
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.000973
```

```
Index: 50
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.001299
```

```
Index: 51
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.002220
```

```
Index: 52
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.000747
```

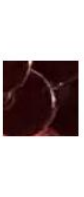
```
Index: 53
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.002536
```

```
Index: 54
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.001538
```

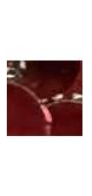
```
Index: 55
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.001324
```

```
Index: 56
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.001324
```

```
Index: 57
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.008696
```



```
Index: 58
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.000368
```



```
Index: 59
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.000610
```

```
Index: 60
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: brown
-- Matched --
INSTANCE: brown
CATEGORY: brown
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.001084
```

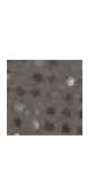
```
Index: 61
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.000413
```

```
Index: 62
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.000368
```

```
Index: 63
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.000610
```

```
Index: 64
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.001608
```

```
Index: 65
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.000967
```



```
Index: 66
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.001925
```

```
Index: 67
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.000967
```

```
Index: 68
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.001456
```

```
Index: 69
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.003454
```

```
Index: 70
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.001142
```

```
Index: 71
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.000123
```

```
Index: 72
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.001142
```

```
Index: 73
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.000123
```

```
Index: 74
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.000864
```